BIO-RAD

Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, adherent

Molecules Electroporated DNA: varies, fibronectin β-gal, genomic DNA, CMUB, etc.

Species Used Human, fibroblast; Human Hep3b2, hepatocyte; Mouse, L-cells.

Before the Pulse

Cell Growth Medium MEM + 10% Fetal Calf Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest Log

Pre-pulse Incubation 5 minutes

Wash Solution Phosphate Buffered Saline and Trypsin

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature 25 °C

Electroporation Medium* MEM

Cell Density 1 to 10 x 10 (6) cells / pulse

Volume of Cells 500 µl

DNA Concentration 40 to100 µg DNA per pulse

DNA Resuspension Buffer TE (10 mM Tris, 1 mM EDTA, pH 8.0)

Volume of DNA 20 to 40 µl

Cuvette Gap 0.4 cm

Voltage 0.320 kV

Field Strength 0.8 kV/cm

Capacitor 500 µF

Resistor (Pulse Controller) Ω none

Time Constant Not given

After the Pulse

Outgrowth Medium MEM

Outgrowth Temperature 37 °C

Length of Incubation 48 hours to 1 month

Selection Method or Assay Used GPT and Neomycin

Electroporation Efficiency 60%

Per Cent Survival 50%

Relevant Publications and/or Comments
Note: exponential values designated in parentheses.

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